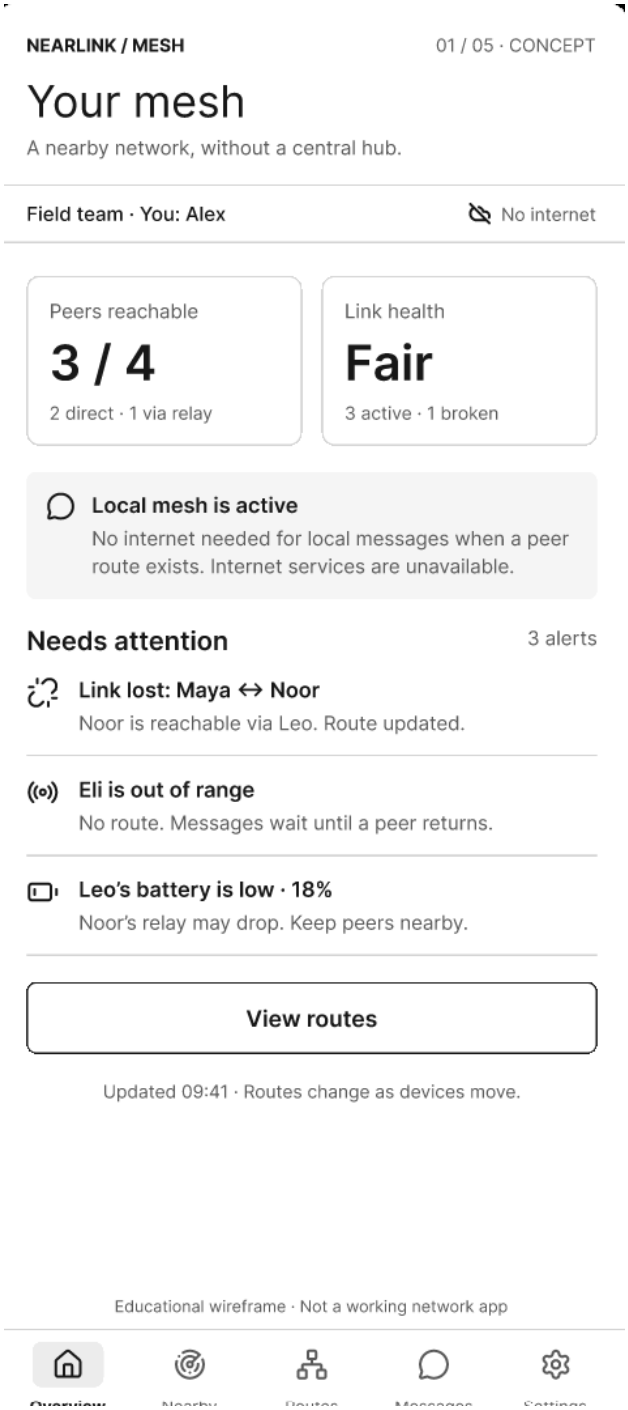
EXPERIMENT 15

Develop a wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma.



NEARLINK / MESH

02 / 05 · CONCEPT

Nearby devices

Discover peers within direct radio range.

Field team · You: Alex

No internet

3 devices found

Discovery on · Every 30 s

Scan now

Maya

Strong · −52 dBm

Trusted · Direct link

Connected

Leo

Fair · −71 dBm

Trusted · Direct link

Connected

Sam

Weak · −84 dBm

Untrusted · Not in mesh

Review & join

Check before you connect

For Sam, compare a pairing code in person before trusting or joining. Weak signals may drop.

Not directly nearby

Noor · Reachable through Leo · Trusted

Eli · Out of range · Trusted · No route

Educational wireframe · Not a working network app

Overview | Nearby | Routes | Messages | Settings

Topology & routes

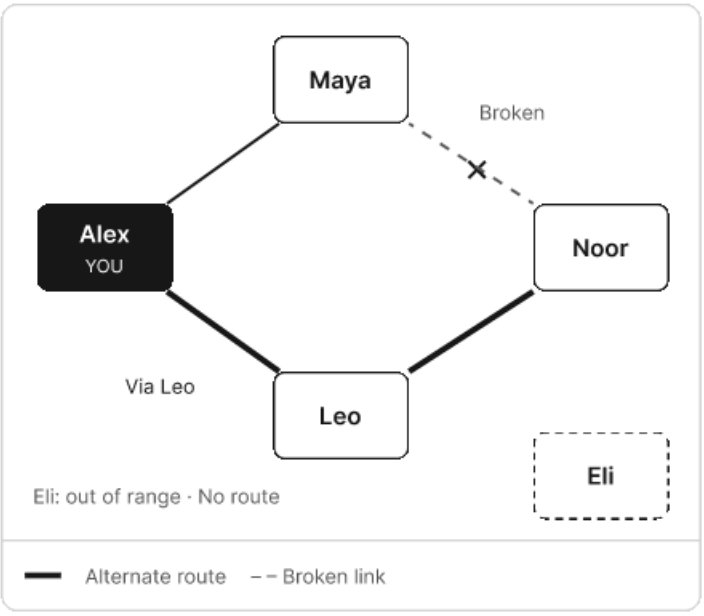
Devices relay messages for one another.

Field team · You: Alex

No internet

Field team · 5 devices

Updated 09:41



Route to Noor restored

Alex → Leo → Noor 2 hops

Maya ↔ Noor failed. The alternate path uses Leo, whose battery is low (18%). All shown peers are trusted.

Routes update as peers move. If every path breaks, messages wait for a route to return.

NEARLINK / MESH

04 / 05 · CONCEPT

Local messages

Send to a peer or your Field team group.

Field team · You: Alex

No internet

Local messages work without internet when a peer route exists. Pending messages stay on this device.

Recent outgoing

Field team · 5 members

To Noor · Direct message

09:40

Meet us at the north gate.

Delivered

Reached Noor via Leo · 2 hops

To Field team · Group

09:41

Bring the map. We're ready to leave.

Queued

Saved on this device · Retry in 10 s

To Eli · Direct message

09:41

Where are you? We'll wait here.

Awaiting route

Eli is out of range · Not delivered

New local message

Queued messages retry automatically. Awaiting-route messages wait until a path is available.

Educational wireframe · Not a working network app

Overview Nearby Routes **Messages** Settings

NEARLINK / MESH

05 / 05 · CONCEPT

Network settings

Choose who can join and how often to scan.

Field team · You: Alex

No internet

Security & trust

Require join approval
Review a peer before allowing access.

Auto-connect trusted peers
Only peers you have already verified.

Trusted devices
Maya, Leo, Noor and Eli. Sam is not trusted.
4 >

Battery & discovery

Your battery: 62%

Battery saver
Scan every 30 s; new peers appear later.

Nearby discovery
Make this device visible to nearby peers.

Relay for other peers
Uses battery; off may break others' routes.

Verify trust in person
Compare a pairing code before trusting a new peer. Remove unfamiliar devices from Trusted devices.

Concept only: connectivity and security need a real implementation. Radio and OS support vary by device.

Educational wireframe · Not a working network app

Overview Nearby Routes Messages Settings